
ABIODUN EMMANUEL ABIOYE

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ENGINEERING AND COMPUTER SCIENCE INSTRUCTOR

CAREER HIGHLIGHTS

Engineering and Computer Science Instructor: Provided expert instructions and supervised practical experiments for students of a Hispanic-serving institution.

Precision Agriculture Researcher: Experience developing predictive models to assess and predict grape composition for Gallo E & J, resulting in over 90% accuracy, and optimizing vineyard robot navigation

SKILLS

Languages	Precision Agric Tech	Data Science	Generative AI	Modelling
C++, HTML, Python, MATLAB	Sensors RGB, Multispectral Hyperspectral Imaging	Pytorch, TensoFlow Scikit-learn OpenCV	LLMs, Chatbots VLMs, Azure, AWS, and GCP	Microsoft Office Machine Learning Deep Learning

EDUCATION

PhD in Electrical Engineering(Precision Agriculture) Nov. 2021	Universiti Teknologi Malaysia
Master's Degree in Electronics and Communication Engineering Oct. 2015	University of Agriculture, Umudike Abia State, Nigeria
Bachelor's Degree in Electrical Engineering July 2010	University of Ilorin, Ilorin Nigeria

PROFESSIONAL MEMBERSHIPS

Member, American Geophysical Union	2025
Member, American Society of Agricultural and Biological Engineers (ASABE)	2022
Member, American Society of Mechanical Engineers	2025
International Society of Precision Agriculture	2021

CERTIFICATION

Cisco Certified Network Associate (CCNA)	2016 - 2019
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VOLUNTEERING

Volunteer	USAID-Winrock International West Africa Farmer To Farmer
- Training of farmers in Guinea-Africa on best practices for solar energy management - Preparation of training manual modules.	

PROFESSIONAL WORK EXPERIENCE

Engineering/Computer Science Instructor	Reedley College	Reedley, California
Aug. 2024– Till now	- Carrying out applied research on Agricultural sensors and social robotics. - Team lead of the Reedley College Ag Robotics Research Team for the Farm Robotic Challenge. - Team member of the Reedley College Dielectric Actuators Research Group. - Teaching CSCI-40- Programming Concepts and Methodology (C++), ENGR 10-Introduction to Engineering, CSCI-1 Introduction to Computer Science (Python), ENGR 5 - Programming and Problem Solving in MATLAB - Conducting practical experiments and research in STEM education (INTDS 101 & INTDS 103) - Supervising students' capstone projects and activities (STEM night and SCI-FRI)	

Postdoctoral Scholar**California State University****Fresno, CA**

March 2023– March 2024

- I developed a predictive model to assess and predict grape composition for Gallo E & J, resulting in over 90% accuracy.
- I developed and implemented predictive models to assess and forecast water status in plants, improving incident response capabilities of plant water stress and improving irrigation efficiency by 30%.
- Collaborated with faculty and research teams to analyze data and generate actionable insights for enhancing irrigation prediction.

Postdoctoral Scholar**University of California****Davis, CA**

March 2022– March 2023

- Lead effort in the use of Internet of Things devices and other instrumentation for field data Collection in orchards and vineyards.
- Data preprocessing and mechanistic modeling
- Co-developed a decision support system for optimal spray application.

RESEARCH GRANT

Consortium of Universities for the Advancement of Hydrologic Science, Inc, Instrumentation Discovery Travel Grant, OPEnS Lab, Oregon State University, 2025

SOME SELECTED PUBLICATIONS

Abioye, E. A., Abidin, M. S. Z., Mahmud, M. S. A., Buyamin, S., Ishak, M. H. I., Rahman, M. K. I. A., Otuoze, A.O., Onotu, P., & Ramli, M. S. A. (2020). A review on monitoring and advanced control strategies for precision irrigation. *Computers and Electronics in Agriculture*, 173, 105441. <https://doi.org/10.1016/j.compag.2020.105441>

Abioye, A.E., Laroche-Pinel, E., Sams, B., Corales, B., Vasquez, K., Cianciola, V. and Brillante, L. (2024). Grape composition assessment using NIR/SWIR hyperspectral imagery acquired from a UTV. *Acta Hort.* 1395, 351-358, DOI: 10.17660/ActaHortic.2024.1395.46 <https://doi.org/10.17660/ActaHortic.2024.1395.46>

Abioye, E. A., Abidin, M. S. Z., Aman, M. N., Mahmud, M. S. A., & Buyamin, S. (2021). A model predictive controller for precision irrigation using discrete Laguerre networks. *Science Direct-Computers and Electronics in Agriculture*, 181, 105953. <https://doi.org/10.1016/j.compag.2020.105953>

Abioye, E. A., Abidin, M. S. Z., Mahmud, M. S. A., Buyamin, S., Ijike, O. D., Otuoze, A. O., Afis, A. A., & Olajide, O. M. (2023). A data-driven Kalman filter-PID controller for fibrous capillary irrigation. *Science Direct-Smart Agricultural Technology*, 3, 100085. <https://doi.org/10.1016/j.atech.2022.100085>

Abioye, A. E., Abidin, M. S. Z., Mahmud, M. S. A., Buyamin, S., Mohammed, O. O., Otuoze, A. O., Oleolo, I. O., & Abioye, M. (2023). Model-based predictive control strategy for water-saving drip irrigation. *Science Direct-Smart Agricultural Technology*, 4, 100179. <https://doi.org/10.1016/j.atech.2023.100179>

Abioye, A. E., & Larbi, P. A. (2023). Assumptions and considerations for a novel mechanistic airblast spray drift model for orchards and vineyards. ASABE Meeting Presentation. <https://doi.org/10.13031/aim.202300224>

Abioye A. E., Zainal Abidin, M. S., Mahmud, M. S. A., Buyamin, S., AbdRahman, M. K. I., Otuoze, A. O., et al. (2020). IoT-based monitoring and data-driven modelling of drip irrigation system for mustard leaf cultivation experiment. *Science Direct-Inf. Proc. in Agriculture*. <https://doi.org/10.1016/j.inpa.2020.05.004>

Abioye, E. A., Hensel, O., Esau, T. J., Elijah, O., Abidin, M. S. Z., Ayobami, A. S., Yerima, O., & Nasirahmadi, A. (2022). Precision Irrigation Management Using Machine Learning and Digital Farming Solutions. *AgriEngineering*, 4(1), 70-103. <https://doi.org/10.3390/agriengineering4010006>

Abioye, A. E., Larbi, P. A., & Hadwan, A. A. K. (2024). Deep learning guided variable rate robotic sprayer prototype. *Smart Agricultural Technology*, 9, 100540. <https://doi.org/10.1016/j.atech.2024.100540>

O. Elijah, **Abioye A. E.** and T. E. Maguvu, "Pest and Disease Management in Ginger Plants: Artificial Intelligence of Things (AIoT)," in *IEEE Transactions on AgriFood Electronics*, vol. 3, no. 1, pp. 86-97, March-April 2025, doi: 10.1109/TAFE.2024.3492323.

O. Elijah, S. K. A. Rahim, **Abioye, A.E.**, Y. O. Salihu, Z. G. Usman and A. M. Jimoh, "Enabling Smart Agriculture in Nigeria: Application of Digital-Twin Technology," *2021 1st International Conference on Multidisciplinary Engineering and Applied Science (ICMEAS)*, Abuja, Nigeria, 2021, pp. 1-6, doi: 10.1109/ICMEAS52683.2021.9692351.

LINKEDIN PROFILE/ RESEARCH PROFILE<https://www.linkedin.com/in/abiodun-e-abioye-11b35386/>https://scholar.google.com/citations?user=Zp_XCJcAAAAJ&hl=en<https://github.com/abioduneabioye>

AI-Powered Decision Support for Precision Agriculture

Developed Python-based machine learning models to predict grape composition and optimize irrigation efficiency during postdoctoral research at California State University, Fresno. Integrated environmental and sensing data to support climate-smart water management and improve agricultural resource efficiency.

Hyperspectral Imaging & Predictive Modeling for Crop Health Assessment

Designed and validated predictive models using NIR/SWIR hyperspectral data to assess fruit composition and crop conditions. This work contributed to early detection strategies and improved monitoring in vineyard systems.

AI-Driven Chatbots and LLM-Based Systems for Information Access

Designed and implemented LLM-powered chatbots for web applications and human–robot interaction (social robotics), focusing on conversational workflows, prompt engineering, and backend API integration to deliver real-time, user-centered support.

Data-Driven Impact Modeling and Analytical Workflows

Applied statistical analysis and data validation techniques to manage complex datasets, document assumptions, and support transparent, reproducible modeling workflows relevant to climate and environmental impact assessment.

Embedded and Systems-Oriented Engineering Projects

Worked with C/C++ and embedded toolchains (e.g., Code Composer Studio) in microcontroller-based projects, supporting system-level understanding of hardware–software integration.

STEM Education and Workforce-Oriented Curriculum Development

Designed and delivered programming and engineering curricula in Python, C++, and MATLAB at the college level. Supervised capstone projects emphasizing applied problem-solving, sustainability, and industry-relevant skills, and explored AI-enabled workflows for adaptive curriculum development.

Renewable Energy and Sustainable Development Field Training

Volunteered with USAID–Winrock International’s Farmer-to-Farmer program in West Africa, delivering training on solar energy management and developing practical manuals to support renewable energy adoption in resource-constrained communities.

REFERENCES

Dr. John Heathcote

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